

Large eddy simulation with direct resolution of subgrid motion using a grid free vortex particle method

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ABSTRACT

The paper presents a novel Large Eddy Simulation approach with a direct resolution of the subgrid motion of fine concentrated vortices. The method, proposed first by Kornev (2018), is based on combination of a grid based and the grid free computational vortex particle (VPM) methods. The large scale flow structures are simulated on the grid whereas the concentrated structures are modeled using VPM. Due to this combination the advantages of both methods are strengthened whereas the disadvantages are diminished. The procedure of the separation of small concentrated vortices from the large scale ones is based on LES filtering idea. The flow dynamics is governed by two coupled transport equations taking two-way interaction between large and fine structures into account. The fine structures are mapped back to the grid if their size grows due to diffusion. Algorithmic aspects specific for three dimensional flow simulations are discussed. Validity and advantages of the new approach are illustrated for a well tried benchmark test of the decaying homogeneous isotropic turbulence using the experimental data of Comte-Bellot and Corrsin (1971).

1. Introduction

Insufficient resolution of fine structures is one of the key problems in Computational Fluid Dynamics (CFD). The most advanced and popular technique to resolve multi scale flow structures is the Large Eddy Simulation (LES) which is based on the idea of scale decomposition into large and small ones. While the large eddies are directly resolved on the grid, the effect of small vortices is taken into account through a subgrid stress (SGS) model. There are many problems which require direct resolution of the subgrid motion to simulate, for instance, mixing or particle dynamics in turbulent flows.

In this paper, we propose a kind of LES version with direct resolution of the subgrid motion. To increase the vortex resolution it is suggested to apply a hybrid grid- and particle based method, utilizing a combination of the finite volume and computational vortex particle (VPM) (see Cottet and Koumoutsakos (2000)) methods. It is based on the decomposition of the velocity and vorticity fields into the distributed large scale and concentrated small scale fields. The large scale field is represented on the grid like in LES, whereas the small scale one is calculated using the VPM. The method is a purely Lagrangian one for small structures and purely grid based one for large scale structures. There exist a permanent exchange and a strong coupling between grid- and particle represented vortices described by Kornev (2018).

For the overview of particle based methods the reader is referred to works of Koumoutsakos (2005), Cottet and Koumoutsakos (2017), Winckelmans et al. (2016), Mimeau (2015) and Stock and Garakhani (2010). An extension of the VPM to multi phase flows can be found in Chen et al. (2014). The computational vortex particle method is far from new one and is still not so popular as other meshless and non-conventional methods like smooth partial hydrodynamics (SPH) and Lattice Boltzmann method (LBM) which made a serious progress in last decades even for engineering tasks because of availability of free codes. Applications of VPM for viscous flows is still restricted to a narrow range of specific problems mostly at small Reynolds numbers. Perhaps, one of the reasons is that computational particle methods generally require tuning in the process of numerical calculations, which is a know-how (Grigoryev et al., 2002) and usually not published. Instabilities and fluctuations of simulations belong to a substantial problem of particle methods, which according to Grigoryev et al. (2002) are connected with comparatively small number of model particles used in calculations. In the author's opinion, the instability of VPM simulations of three dimensional flows with concentrated vortices at high Reynolds numbers is due to insufficient resolution of the vorticity enhancement and vorticity diffusion as two competitive physical processes determining the vortex dynamics (see also Section 3.3).

One of the most important trends in modern VPM versions is their

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coupling with grid based methods. While a part of vortex method experts continue to use the traditional vortex method in a purely Lagrangian form, other groups consider the vortex method only in combination with grid based simulations using the Vortex in Cell (VIC) approach and remeshing. In a typical splitting scheme presented, for instance by Mimeo (2015), calculation of velocity from Poisson equation, fulfillment of the no slip boundary condition through the Brinkman penalization method and transport of vorticity are calculated on the grid. Only the convection of vortex elements is treated in a purely Lagrangian way. Simple uniform Cartesian grids are commonly utilized because of application of the Fast Fourier Transformation procedures and high order interpolation of flow parameters from grid to particles and back. Like in LBM, refinement can be done by overlapping of Cartesian grids (see El Osmani and Poncet (2010)). Involvement of the grids provide a stable and accurate solution at least at low Reynolds numbers. However, the loss of the most important advantages of pure Lagrangian methods, i.e. grid independence, raises the big question about the efficiency and competitiveness of VPM method with respect to common grid based techniques which were brought to perfection by use of massive parallelism of computations.

Despite the shortcomings mentioned above, VPM remains very attractive technique because of reduced numerical viscosity especially within purely Lagrangian formulations. It could also be very efficient for solution of special problems with rapidly changing complex flow boundaries. In this paper, the VPM is used in combination with grid based method to simulate the subgrid motion which is not resolved on the grid. Thus, the VPM plays a supporting role of the subgrid model. Since we use the formalism different from that of the classical VPM method many of VPM weaknesses become irrelevant.

Our method is based on the assumption that small scale effects in turbulent flows are determined by a countable number of fine concentrated vortices which are approximately axisymmetric. Subgrid vortices are represented by simple particles with axisymmetric distribution of vorticity (30). The particles used in particle methods are mathematical objects rather than real prototypes even in purely Lagrangian methods. The particles have size, shape and inner distribution of physical quantity carried by the particle. Their choice is dictated by the wish to simplify the calculations of particle dynamics. We believe that the particles (30) are not so far from their real prototypes from two reasons. First, measurements show that the fine scale vorticity is very intermittent and discrete in its nature and can be modeled by discrete particles. Second, as shown in DNS simulations, the fine scale vorticity is organized in thin tubes with the resulting vortex core in form of an approximate solid-body rotation. More precisely the thickness of these tubes is intermediate between the dissipation scale and the Taylor microscale, while their length is comparable with the integral scale of the flow (see, for instance, Vincent and Meneguzzi (1994)). Therefore, approximation of such tubes with axisymmetric particles can be considered as a proper one.

At high Reynolds numbers the small vortices have scales so small that any exact reproduction of all of them using any grid connected technique would be a very hard task. Extremely fine grid should cover the whole domain including regions which are at the moment not occupied by vortices, since they can arise here at the next time moment. A realizable method should be designed in such a way that only certain effects have to be reproduced in a statistical sense. This seems to be possible if only the most energetic fine vortices are taken into account. The success of such a method should thoroughly be validated for certain well tried benchmark tests.

In this paper the method is validated for the decaying homogeneous isotropic turbulence case using the experimental data of Comte-Bellot and Corrsin (1971). At present there are at least two VPM simulations of the decaying homogeneous isotropic turbulence. Cottet et al. (2002) utilizes the vortex-in-cell (VIC) which uses the grids for computations of the diffusion, stretching and velocity calculations, leaving only the convection to be calculated using a Lagrangian

approach. Yokota et al. (2007) applied a pure Lagrangian method which, however, is combined with grid based procedures like remeshing and Fast Multipole Method. Similarly to the present paper, Yokota et al. (2007) used the smoothing length as a physical core radius of the vortex blob growing according to the diffusion equation. Initial conditions are generated in Yokota et al. (2007) for the model spectrum $E(k) = k^4 e^{-2k^2/k_p^2}$ which according to our experience is very far from that measured in Comte-Bellot and Corrsin (1971). In contrast to the present paper, Cottet et al. (2002) and Yokota et al. (2007) evaluate the quality of vortex method simulations using again the simulation, performed with a spectral method.

The method, most similar to the present one, was proposed in Selle et al. (2005). The difference is that Selle et al. (2005) consider the fine vortices as a passive phase driven by large scale motion without their interaction. Since the VPM simulation within the present method is embedded into the grid simulation, the method is referred to as the hybrid embedded simulation to distinguish it from the hybrid methods of Ould-Salihi et al. (2000) and Feder et al. (2018) based on the domain decomposition.

2. Governing equations

The flow is assumed to be isothermal and incompressible. The governing equations are derived using the decomposition of the velocity and vorticity fields into large scale, represented on the grid, and fine scale, represented by vortex particles, components $\mathbf{u} = \mathbf{u}^g + \mathbf{u}^v$ and $\boldsymbol{\omega} = \boldsymbol{\omega}^g + \boldsymbol{\omega}^v$. Substitution of the components into the vorticity transport equation gives:

$$\frac{\partial(\boldsymbol{\omega}^v + \boldsymbol{\omega}^g)}{\partial t} + ((\mathbf{u}^v + \mathbf{u}^g) \cdot \nabla)(\boldsymbol{\omega}^v + \boldsymbol{\omega}^g) = ((\boldsymbol{\omega}^v + \boldsymbol{\omega}^g) \cdot \nabla)(\mathbf{u}^v + \mathbf{u}^g) + \nu \Delta(\boldsymbol{\omega}^g + \boldsymbol{\omega}^v) \quad (1)$$

which can be splitted into two following equations:

$$\begin{cases} \frac{\partial \boldsymbol{\omega}^g}{\partial t} + ((\mathbf{u}^v + \mathbf{u}^g) \cdot \nabla) \boldsymbol{\omega}^g = (\boldsymbol{\omega}^g \cdot \nabla)(\mathbf{u}^v + \mathbf{u}^g) + \nu \Delta \boldsymbol{\omega}^g \\ \frac{\partial \boldsymbol{\omega}^v}{\partial t} + ((\mathbf{u}^v + \mathbf{u}^g) \cdot \nabla) \boldsymbol{\omega}^v = (\boldsymbol{\omega}^v \cdot \nabla)(\mathbf{u}^v + \mathbf{u}^g) + \nu \Delta \boldsymbol{\omega}^v \end{cases} \quad (2)$$

Using identities of the vector analysis one gets the $u - p$ formulation for the first equation of (2):

$$\frac{\partial \mathbf{u}^g}{\partial t} + (\mathbf{u}^g \cdot \nabla) \mathbf{u}^g = -\frac{1}{\rho} \nabla P^g + \nu \Delta \mathbf{u}^g + (\mathbf{u}^v \times \boldsymbol{\omega}^g) \quad (3)$$

$$\frac{d\boldsymbol{\omega}^v}{dt} = (\boldsymbol{\omega}^v \cdot \nabla)(\mathbf{u}^v + \mathbf{u}^g) + \nu \Delta \boldsymbol{\omega}^v \quad (4)$$

Two equations are coupled. The first Eq. (3) is coupled with the second one (4) through the additional term $(\mathbf{u}^v \times \boldsymbol{\omega}^g)$ whereas the coupling of the second equation with the first one is due to the terms $(\mathbf{u}^g \cdot \nabla) \boldsymbol{\omega}^v$ and $(\boldsymbol{\omega}^v \cdot \nabla) \mathbf{u}^g$. The Eqs. (3) and (4) are solved sequentially. The first equation is solved on the grid whereas the second one using the grid free computational vortex method. The sum of these two equations written in the same variables retrieves the original Navier–Stokes equation.

The effect of small structures on grid based solution is taken by the term $\mathbf{u}^v \times \boldsymbol{\omega}^g$ into account. The physical meaning of this term can easily be explained when applying the curl operator:

$$\nabla \times (\mathbf{u}^v \times \boldsymbol{\omega}^g) = -(\mathbf{u}^v \cdot \nabla) \boldsymbol{\omega}^g + (\boldsymbol{\omega}^g \cdot \nabla) \mathbf{u}^v \quad (5)$$

The first term on the r.h.s. of (5) describes the transport of the grid based vorticity $\boldsymbol{\omega}^g$ by the velocity induced by small vortices $\mathbf{u}^v$, whereas the second term is responsible for the rotation and amplification of the grid based vorticity in field of $\mathbf{u}^v$. Since the vortices are getting small due to stretching of vortex lines and can become invisible on the grid, this term needs a special treatment. The term $\mathbf{u}^v \times \boldsymbol{\omega}^g$ involves spatial and temporal scales which can be much less than the quantities represented on the grid. This can lead to the stiffness of the equation (3) which is a common problem in modeling of multiscale physics. An usual

way to overcome this problem is the averaging of terms representing fine scales (see Weinan (2011)). Generally, the term $\mathbf{u}^v \times \omega^g$ can produce scales compared with cell size and beyond. To escape the stiffness the Eqs. (3) and (4) are rewritten in the form:

$$\frac{\partial \mathbf{u}^g}{\partial t} + (\mathbf{u}^g \cdot \nabla) \mathbf{u}^g = -\frac{1}{\rho} \nabla P^g + \nu \Delta \mathbf{u}^g + \overline{\mathbf{u}^v \times \omega^g} \quad (6)$$

$$\frac{d\omega^v}{dt} = (\omega^v \cdot \nabla)(\mathbf{u}^v + \mathbf{u}^g) + \nu \Delta \omega^v + \nabla \times [\mathbf{u}^v \times \omega^g - \overline{\mathbf{u}^v \times \omega^g}], \quad (7)$$

where $\overline{\mathbf{u}^v \times \omega^g}$ is the spatially filtered vector. The continuity equations should be satisfied for velocities $\mathbf{u}^g$, $\mathbf{u}^v$ and vorticities ω^g and ω^v . Following the idea of the VPM method the fine vorticity ω^v is represented by vortex particles with strengths expressed through the grid based velocity $\mathbf{u}^g$ utilizing the algorithms described in the next section. The velocity $\mathbf{u}^v$ is then calculated through ω^v using the Bio-Savart law. Hereby the system (6) and (7) becomes closed.

Averaged terms are often used in transport equations describing multiscale processes to avoid stiffness. For instance, averaged source term is used in species transport equation in combustion, subgrid stresses in large eddy simulation (LES) equations, Reynolds stresses in RANS equations, etc. They cannot be calculated from transport equations for averaged quantities. They should be modeled and non-universality of these closure models is the biggest weakness of many theories. An important advantage of our approach is that $\mathbf{u}^v \times \omega^g$ can be calculated without any closure models because $\mathbf{u}^v$ and ω^g are both known. Averaging is necessary to avoid numerical noise caused by irregular distribution of vortex particles inside of cells. Our formalism does not use commutation of differencing and filtering operators which is the big difficulty in LES formalism.

The decomposition (3) and (4) is non-unambiguous. For instance, gathering all grid based quantities in the first equation one obtains the system:

$$\frac{\partial \mathbf{u}^g}{\partial t} + (\mathbf{u}^g \cdot \nabla) \mathbf{u}^g = -\frac{1}{\rho} \nabla P^g + \nu \Delta \mathbf{u}^g + (\mathbf{u}^v \times \omega^g) + (\mathbf{u}^g \times \omega^v) \quad (8)$$

$$\frac{\partial \omega^v}{\partial t} + (\mathbf{u}^v \cdot \nabla) \omega^v = (\omega^v \cdot \nabla) \mathbf{u}^v + \nu \Delta \omega^v \quad (9)$$

with the second equation being independent of the first one. With the use of identity $(\mathbf{u}^v \times \omega^g) + (\mathbf{u}^g \times \omega^v) = \nabla(\mathbf{u}^g \mathbf{u}^v) - (\mathbf{u}^g \cdot \nabla) \mathbf{u}^v - (\mathbf{u}^v \cdot \nabla) \mathbf{u}^g$ the first equation takes the form

$$\frac{\partial u_i^g}{\partial t} + \frac{\partial u_i^g u_j^g}{\partial x_j} = \frac{\partial}{\partial x_i} \left(\frac{P^g}{\rho} + u_k^g u_k^v \right) + \frac{\partial}{\partial x_j} \left(\nu \frac{\partial u_i^g}{\partial x_j} - (u_i^g u_j^v + u_i^v u_j^g) \right) \quad (10)$$

The term $u_i^g u_j^v + u_i^v u_j^g$ is the stress caused by fine scale motion. The term $u_k^g u_k^v$ is a sort of source term or a part of the normal stress. The choice of the proper decomposition can be done on the base of computational experience and still remains open. In this paper we use the formulation (6) and (7) because of clear physical interpretation of interaction terms in both equations.

The large scale flow field represented on the grid is separated from the small scale one to be represented by vortex particles using the filtration procedure taken from Large Eddy Simulation (LES). First, the original velocity field $\mathbf{u}^g$ is filtered with some filter function $F(\mathbf{x} - \mathbf{s})$:

$$\overline{\mathbf{u}}^g(\mathbf{x}, t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{u}^g(\mathbf{s}, t) F(\mathbf{x} - \mathbf{s}) d\mathbf{s} \quad (11)$$

The small scale velocity field calculated as the difference between the original and filtered fields should be approximated by vortex particles.

$$\mathbf{u}^v(\mathbf{x}, t) = \mathbf{u}^g(\mathbf{x}, t) - \overline{\mathbf{u}}^g(\mathbf{x}, t) \quad (12)$$

The vortex particles are placed only in regions of concentrated vortices which are detected using any vortex identification criteria, for instance, λ_{ci} (Chakraborty et al. (2005)). The algorithm of approximation is

described in the next subsection 7. The fine and large scales are strongly interacting according to mechanisms recapitulated in the chapter 4 in Kornev (2018).

3. Numerical implementation

3.1. Identification and generation of vortex particles within the grid based velocity field

Creation of small scale vortices occurs in the large scale flow due to natural instability. As described in Kornev (2018), the first step of vortex identification is the computation of the pulsation velocity field as the difference between the original and filtered fields at each grid point (see (12)). Then the vortex identification criterion λ_{ci} is calculated at each cell generating the field λ_i . The cells at which $\lambda_i > \lambda_{ci, min}$ contain the vortices which in principle can be converted to vortex particles. They will be marked as active ones using the $\lambda_{i, active}$ field:

$$\lambda_{i, active} = \begin{cases} 1, & \text{if } \lambda_{ci} > \lambda_{ci, min} \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

where $\lambda_{ci, min}$ is a certain small value introduced to sort out the weak vortices. To keep the required computational resources on an acceptable level, only small vortices with size proportional to the local cell size Δ are to be converted to single vortex particles. It is supposed that the larger vortices with scales of a few Δ can accurately be represented on the grid. Therefore they should be identified and separated from small scale ones. In the paper Kornev (2018) we have already proposed a procedure for identification of fine vortex structures which are replaced by vortex particles if the size of the structures is comparable with the cell size. The procedure is based on the sequential identification of points with the maximum vorticity and the vortex tubes going through these points. This procedure is proved to be very computational time consuming and can not be used for large three dimensional computations on grids with a few dozens of millions of cells.

Based on this experience, a more robust and fast approach was developed in this paper, which doesn't use neither the vortex tube concept, which identification is a rather complex problem, nor connectivity property of vortex particles. According to the algorithm, all neighboring cells of the i -th cell are checked for the condition $\lambda_{ci} > \lambda_{ci, min}$. If all neighbors fulfill this condition we identify a cell cluster which remains on the grid and all their cells become non-active $\lambda_{i, active} = 0$. Only activated vortices in cells with $\lambda_{i, active} = 1$ are to be replaced by vortex particles. At the present stage of our development, the cluster identification for the unstructured grids implemented in OpenFoam is performed only for cells with joint faces marked by A in Fig. 1. The neighbors marked by B are converted to vortex particles. This means that the number of particles is a little bit larger than it could be. In the future, this shortage can be overcome by enlargement of the

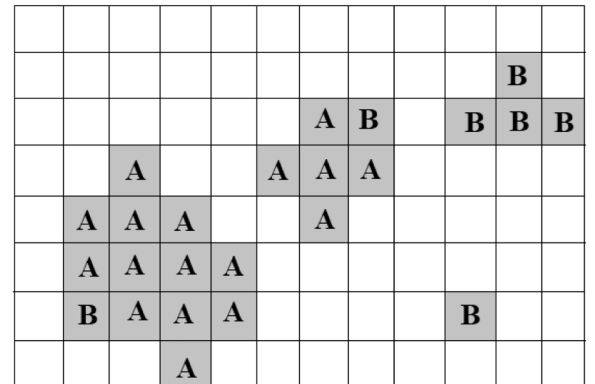


Fig. 1. Cells with $\lambda_{ci} > 0$ (shadowed), cluster cells (A), cells with fine scale vortices converted to vortex particles (B).

neighbor list, which can be pre-processed in OpenFoam if it will be necessary. Thus, the vortices with scales of less than three cell sizes and some vortices located on clusters borders are converted to particles. At each cell with $\lambda_{i,active} = 1$ a single new vortex particle is introduced at the cell center. Thus, each cell contains old vortices and the new one. If the number of vortices is less than N_{pt} no vortices are deleted. Otherwise, the weakest vortices populating this cell are deleted keeping the total number of vortices per cell less or equal to N_{pt} . The radius of the new vortex is set as $\sigma = \beta Vol_i^{1/3}$, where β is the overlapping ratio which is taken as $\beta = 2$ and Vol_i is the volume of the i -th cell. The vortex particle strength is calculated as $\alpha = Vol_i \nabla \times \mathbf{u}'$. If the distance from the new particle to one already existing in the cell is smaller than some permissible distance $\alpha\Delta$, $\alpha \ll 1$ they are merged.

The velocity $\mathbf{u}^v(\mathbf{x}, t)$, induced by the vortex particle, is calculated at grid points using the Bio-Savart law and subtracted from the grid velocity $\mathbf{u}^{g,new} = \mathbf{u}^g - \mathbf{u}^v$. Thus, the total velocity at grid points $\mathbf{u}^{g,new} + \mathbf{u}^v = \mathbf{u}^g$ remains constant after the vortex particle generation procedure.

3.2. Calculation of velocity induced by vortex particles

Computation of the velocity induced by vortex particles is an acknowledged bottleneck of vortex particle methods. Direct summation using Bio-Savart law is not applicable because the number of required operations is proportional to $O(N^2)$, where N is the particle number. The problem can be reduced using either the Vortex in Cell (VIC) approach or the Fast Multipole Method (FMM). VIC has several limitations. First, common VIC application is based on simple Cartesian grids providing a sufficiently acceleration of computations due to employment of the Poisson equation and Fast Fourier Transformation (FFT). The biggest weakness of the Cartesian grids is the inability of an efficient local refinement to handle flows with complex geometries. The use of many overlapping Cartesian grids is, certainly, a not proper way. Extension of the VIC towards arbitrary grids needs development of procedures of vortex redistribution on unstructured grids which is a complicated problem for high order interpolations. Moreover, the resolution of the VIC method is determined by the resolution of the grid used to solve the Poisson equation. In this case the advantages of the vortex methods become questionable. The same limitations can be ascribed to FMM.

To choose the optimal algorithm, we performed a kind of a-priori analysis done for the turbulent field behind the full-bottomed ship JBC which is a new benchmark test designed by the National Maritime Research Institute in Tokyo to validate CFD in ship hydromechanics. This test case was selected for the analysis since the authors obtained very good results for the prediction of the turbulent kinetic energy in the stern area of JBC (Kornev and Abbas, 2018) using grid based finite volume method and hybrid URANS/LES turbulence model. Therefore, it can be assumed that the vortex structures were also predicted with a good accuracy and can be taken for the analysis. The computations were performed with OpenFOAM code. The numerical setup and grids are given in Kornev and Abbas (2018). This analysis has been carried out using simulations on the complex unstructured grid with 30M cells.

For analysis we take the cells lying outside of the boundary layer within the range $x_m < x < x_s$, where x is the longitudinal coordinate, x_m is the section of the midship frame and x_s is the aft extremity of the ship. The velocity fields calculated from OpenFOAM are processed at each time instant. First, the filtered field $\tilde{\mathbf{u}}^g$ is obtained from the original $\mathbf{u}^g$ one by applying the top hat filter. The difference $\mathbf{u}' = \mathbf{u}^g - \tilde{\mathbf{u}}^g$ is a small scale velocity field which should be approximated by vortex elements. For that the vorticity identification criterion λ_2 is calculated from $\mathbf{u}'$ at each cell and the maximum $\lambda_{2,max}$ is sought. To exclude weak insignificant vortices, only cells at which $\lambda_2 > 0.05\lambda_{2,max}$ are selected for the further analysis. The small scale vortices are located at the cell centers and possess the strength $\alpha_i = \omega' Vol_i$, where $\omega' = \nabla \times \mathbf{u}'$ is the small scale vorticity, Vol is the cell volume and i is the cell number, $i = 1, M$. The velocity induced at point $\mathbf{r}_i$ and calculated using the Biot-

Savart law:

$$\tilde{\mathbf{u}}'_i(\mathbf{r}_i) = \frac{1}{4\pi} \sum_{j=1}^M \frac{\alpha_j \times \mathbf{r}_{ij}}{r_{ij}^3} \quad (14)$$

is an approximation of the field $\mathbf{u}'$. In what follows, the tilde over u means that u is calculated from the Bio-Savart law. For the further analysis we introduce the vortex particles $\alpha_i^g = (\nabla \times \mathbf{u}^g) Vol_i$ and $\tilde{\alpha}_i^g = (\nabla \times \tilde{\mathbf{u}}^g) Vol_i$. The velocities $\tilde{\mathbf{u}}^g$ and $\tilde{\mathbf{u}}^g$ induced by these particles are approximations of the original $\mathbf{u}^g$ and filtered $\tilde{\mathbf{u}}^g$ velocity fields. It should be noted that our implementation of the vortex method is different from what has been previously. The previous versions of the vortex methods were aimed at simulation of large scale coherent structures as a rule at practically not relevant low Reynolds numbers. In those methods velocity of the flow is fully determined by vortex particles. In our implementation the velocity is the sum of the grid fraction $\mathbf{u}^g$ and velocity induced by particles $\mathbf{u}'$ which is a fraction of $\mathbf{u}'$. The first one takes the effect of large coherent structures into account whereas the second one is responsible for small scale vortices. As is known, the small scale vortices are isotropic, i.e. they have no preferential direction and their direction is distributed uniformly around the sphere. With the other words, the correlation between the small scale vortices is weak. From this consideration and from the fact that the small scale vortices are well separated one can assume that the velocities $\mathbf{u}'$ and, consequently, $\mathbf{u}^v$ are much less than $\mathbf{u}^g$. Fig. 2 confirms this assumption. The probability density function of the ratio $|\tilde{\mathbf{u}}'|/|\tilde{\mathbf{u}}^g|$, calculated at all M vortex particles for two time steps, shows that the velocity $\tilde{\mathbf{u}}'$ is in average less than one percent of $\tilde{\mathbf{u}}^g$. The maximum possible ratio is barely about ten percent. Thus, $\tilde{\mathbf{u}}^g$ is the dominating part of the flow velocity and $\tilde{\mathbf{u}}'$, in principle, can be neglected as it is done in Selle et al. (2005). In this case the cloud of vortex particles moves driven only by $\tilde{\mathbf{u}}^g$.

In our algorithm we keep $\mathbf{u}^v$ to capture the interaction between neighboring vortices. To clear the question how wide should be the circle of neighbors, the velocity $\tilde{\mathbf{u}}'$ is calculated using the Biot-Savart law with the introduction of the simple cut-off rule:

$$\tilde{\mathbf{u}}'_{i,cut}(\mathbf{r}_i) = \sum_{j=1}^M \delta \mathbf{u}_{ij}, \quad \delta \mathbf{u}_{ij} = \begin{cases} \frac{1}{4\pi} \frac{\alpha_j \times \mathbf{r}_{ij}}{r_{ij}^3} & \text{if } r_{ij} \leq L(Vol_i^{1/3} + Vol_j^{1/3}) \\ 0 & \text{if } r_{ij} > L(Vol_i^{1/3} + Vol_j^{1/3}) \end{cases} \quad (15)$$

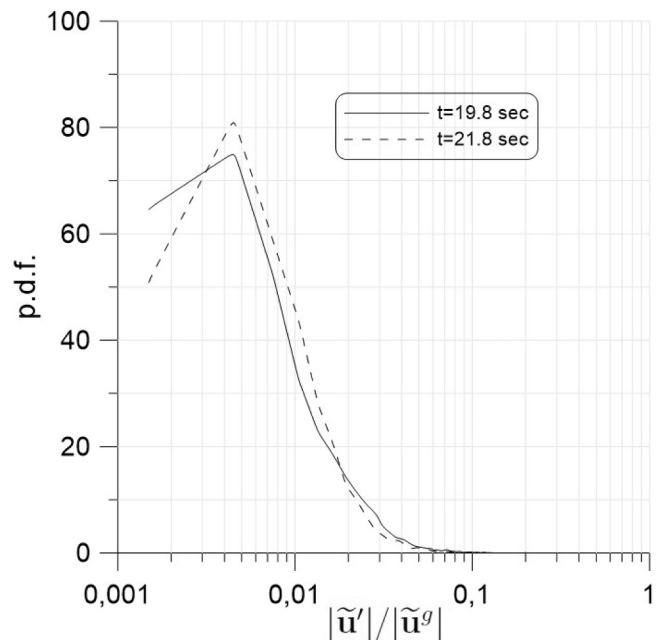


Fig. 2. The probability density function of the ratio $|\tilde{\mathbf{u}}'|/|\tilde{\mathbf{u}}^g|$ at two time instants.

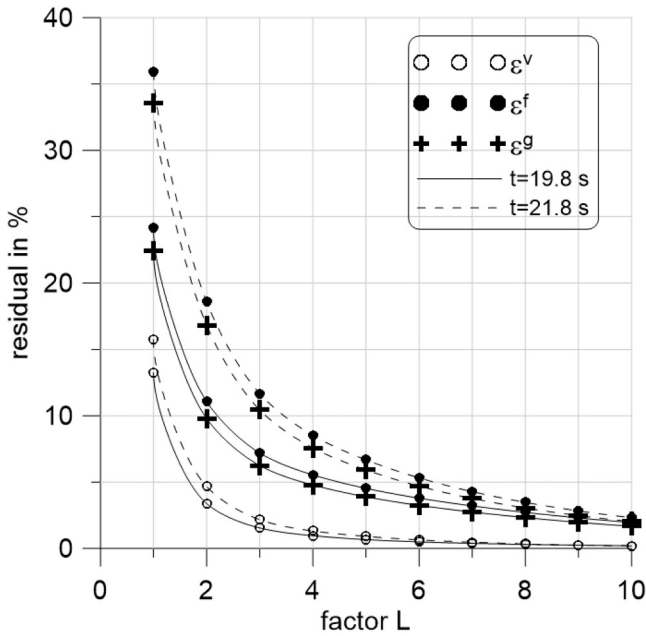


Fig. 3. Residuals ε^v , ε^f and ε^g versus L at two time instants.

Three residuals:

$$\varepsilon^v = \frac{\sum_i^M |\tilde{\mathbf{u}}'_i - \tilde{\mathbf{u}}'_{i,cut}|^2}{\sum_i^M |\tilde{\mathbf{u}}'_i|^2}, \quad \varepsilon^f = \frac{\sum_i^M |\tilde{\mathbf{u}}_i^g - \tilde{\mathbf{u}}_{i,cut}^g|^2}{\sum_i^M |\tilde{\mathbf{u}}_i^g|^2}, \quad \varepsilon^g = \frac{\sum_i^M |\tilde{\mathbf{u}}_i^g - \tilde{\mathbf{u}}_{i,cut}^g|^2}{\sum_i^M |\tilde{\mathbf{u}}_i^g|^2}, \quad (16)$$

where

$$\tilde{\mathbf{u}}_i^g(\mathbf{r}_i) = \frac{1}{4\pi} \sum_{j=1}^M \frac{\alpha_j^g \times \mathbf{r}_{ij}}{r_{ij}^3}, \quad \tilde{\mathbf{u}}_i^g(\mathbf{r}_i) = \frac{1}{4\pi} \sum_{j=1}^M \frac{\bar{\alpha}_j^g \times \mathbf{r}_{ij}}{r_{ij}^3},$$

are presented in Fig. 3 versus parameter L . Increase of L means that more neighboring points are taken into account for calculation of velocity. The residual ε^v is less than five percent already at $L = 2$ whereas other two residuals ε^f and ε^g are over ten percent. Bearing in mind that the same points are used for calculation of all three residuals and they don't depend on the magnitude of velocities $\tilde{\mathbf{u}}^g$, $\tilde{\mathbf{u}}^g$ and $\tilde{\mathbf{u}}'$, the only reason of the fast convergence for small scale induced velocity $\tilde{\mathbf{u}}'$ is a low correlation of small scale vortices ω' . On the contrary, the vortices $\tilde{\omega}^g = \nabla \times \tilde{\mathbf{u}}^g$ and $\omega^g = \nabla \times \mathbf{u}^g$ are more correlated and decay of residual ε^f and ε^g is weaker. The contribution of small scale vortices should also be analyzed for calculation of the strain rate tensor $\partial u_i^v / \partial x_j$ which is necessary for determination of terms $(\omega^v \cdot \nabla) \mathbf{u}^v$ in the vorticity transport equation (7). Vortex amplification rate which is derived from (7) is calculated by multiplication with ω^v :

$$\omega^v \frac{d\omega^v}{dt} = \frac{1}{2} \frac{d\omega^v \omega^v}{dt} = \omega^v ((\omega^v \cdot \nabla)(\mathbf{u}^v + \mathbf{u}^g)) \quad (17)$$

If $\frac{d\omega^v \omega^v}{dt} > 0$ the vorticity magnitude is enhanced, i.e. the small scale vorticity is amplified. Similarly to the above, the analysis is done for $\mathbf{u}'$ and ω' instead of $\mathbf{u}^v$ and ω^v to avoid the influence of errors introduced by the vortex particles generation procedure. Probability density function of the factors of small scale vorticity amplification due to small scale motion $A^v = \omega'((\omega' \cdot \nabla) \tilde{\mathbf{u}}')$ and due to large scale motion $A^g = \omega'((\omega' \cdot \nabla) \tilde{\mathbf{u}}^g)$ as well as due to filtered motion $A^f = \omega'((\omega' \cdot \nabla) \tilde{\mathbf{u}}^g)$ is presented in Fig. 4. Since the p.d.f distribution of A^g is skewed towards positive amplification rate, the small scale vorticity ω' , in average, has tendency to increase. This increase is mostly due to enhancement of A^g caused by large scale motion whereas the contribution of fine scales A^v is small. Since the p.d.f of A^v is symmetric with respect to zero, the probability of enhancement and suppress of small scale vortices caused

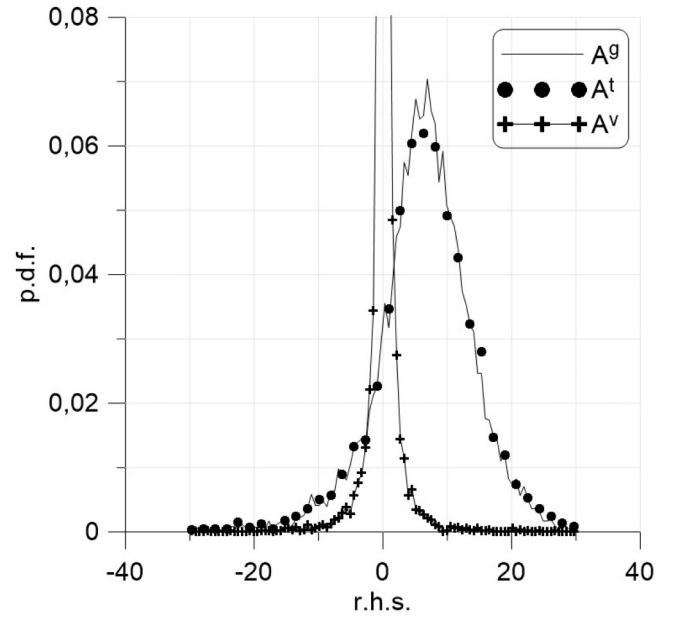


Fig. 4. Right hand side of (17): Amplification factors A^g , A^f and A^v (clipped).

by neighbouring small scale vortices is nearly the same and is zero in average. It should be noted that the p.d.f. of A^g , A^f and A^v is the same for all analyzed time instants.

Based on this analysis it was proposed to calculate $\mathbf{u}^v$ and $(\omega^v \cdot \nabla) \mathbf{u}^v$ with the Bio-Savart law taking into account one or two layers of neighboring cells. This simplification can be considered as a kind of model which results in a very fast computational procedure of local character suitable for parallel calculations.

3.3. Numerical solution of the vorticity transport equation

This problem not addressed in Kornev (2018) is, perhaps, the most important problems of VPM because of the numerical instability caused by the stretching term $(\omega \cdot \nabla) \mathbf{u}$ which is not present in two dimensional flows. In grid based methods with low and moderate order schemes the action of the stretching is effectively counterbalanced by the numerical viscosity which is very low in Lagrangian particle methods. The instability caused by this term is much stronger than that due to irregular distribution of particles. The reader can easily prove it writing a simple code of vortex particle dynamics with and without vorticity strength change. As shown in a classical measurement by Taylor (1938), the stretching is counterbalanced by viscous diffusion. Therefore, theoretically, a stable solution can be obtained by increase of the accuracy of the stretching and diffusion simulation which can be attained by high number of vortex particles and high temporal resolution. Both makes the method impractical at least for high Reynolds numbers. Another way is to use different algorithms containing inherent stabilization like remeshing (see Cottet and Koumoutsakos (2000)) which became inevitable part of the VPM. The remeshing is based on the grid application and possesses the effective resolution proportional to the cell size. The advantages of such a method in comparison with a common grid based methods are questionable and are still being a point of active discussions among VPM experts.

After many efforts, literature recherche and discussions with VPM experts, the authors settled on the scheme which was proposed by Fukuda and Kamemoto (2005) and is based on three conservation laws:

- the conservation of the vortex particle volume
- $$\sigma^2 l = \text{const} \quad (18)$$

where σ and l are the core radius and length of the particle

respectively. From (18) it follows

$$\frac{d\sigma}{dt} = -\frac{\sigma}{2l} \frac{dl}{dt} \quad (19)$$

- the conservation of the circulation of the vortex tube if the viscosity is neglected

$$\sigma^2 |\omega^v| = \text{const} \quad (20)$$

From (20) it follows

$$\frac{d\sigma}{dt} = -\frac{\sigma}{2|\omega^v|} \frac{d|\omega^v|}{dt} \quad (21)$$

and

$$|\omega^v(t + \Delta t)| = |\omega^v(t)| \left(\frac{\sigma(t)}{\sigma(t + \Delta t)} \right)^2 \quad (22)$$

Multiplying (22) with the volume Vol ascribed to the vortex particle we obtain a simple equation for the magnitude of the vortex particle strength α

$$|\alpha(t + \Delta t)| = |\alpha(t)| \left(\frac{\sigma(t)}{\sigma(t + \Delta t)} \right)^2 \frac{Vol(t + \Delta t)}{Vol(t)} \quad (23)$$

Since the volume is proportional to σ^3 the last equation can be written in the form

$$|\alpha(t + \Delta t)| = |\alpha(t)| \frac{\sigma(t + \Delta t)}{\sigma(t)} \quad (24)$$

Stabilizing property of the scheme (Fukuda and Kamemoto, 2005) becomes obvious. In case of stretching $\frac{\sigma(t + \Delta t)}{\sigma(t)} < 1$ the strength of the particle decreases $\frac{|\alpha(t + \Delta t)|}{|\alpha(t)|} < 1$. To avoid the loss of the total vorticity the authors proposed to insert new vortex particles using the redistribution algorithm. This procedure simulates the vortex tube elongation described by the equation derived from (19) and (20):

$$\frac{dl}{dt} = \frac{l}{|\omega^v|} \frac{d|\omega^v|}{dt} \quad (25)$$

The algorithm consists of the following substeps:

- calculation of the change of the vorticity strength magnitude

$$\frac{d|\omega^v|}{dt} = \frac{\omega^v}{|\omega^v|} \frac{d\omega^v}{dt} \quad (26)$$

where $\frac{d\omega^v}{dt}$ is calculated from (7) without viscous diffusion term. The term $(\omega^v \cdot \nabla) \mathbf{u}^v$ is calculated taking into account adjacent vortex particles located only within one or two layers of neighboring cells.

- Calculation of the particle length from (25)
- Calculation of the particle core radius. First, the intermediate core radius σ^* is calculated from (21). The viscosity influence is modeled additionally by the core spreading method (CSM) (Cottet and Koumoutsakos (2000)). Thus, the updated radius is $\sigma(t + \Delta t) = \sigma^* + \sqrt{4\nu\Delta t}$. In contrast to, for instance Particle Strength Exchange (PSE) (Cottet and Koumoutsakos (2000)), CSM is the most suitable way within the general design of our method. If the vortices size grows due to viscosity and flow stagnation they are mapped back to the grid.
- Calculation of the new particle orientation

$$\omega^* = \omega^v(t) + \frac{d\omega^v}{dt} \Delta t \quad (27)$$

- Calculation of the particle strength magnitude from (24)
- Calculation of the new strength vector

$$\alpha(t + \Delta t) = |\alpha(t + \Delta t)| \frac{\omega^*}{|\omega^*|} \quad (28)$$

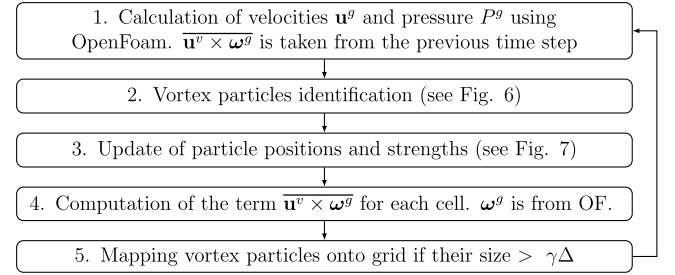


Fig. 5. General structure of the time loop. Steps 2–5 are additional routines implemented into OpenFoam (OF)..

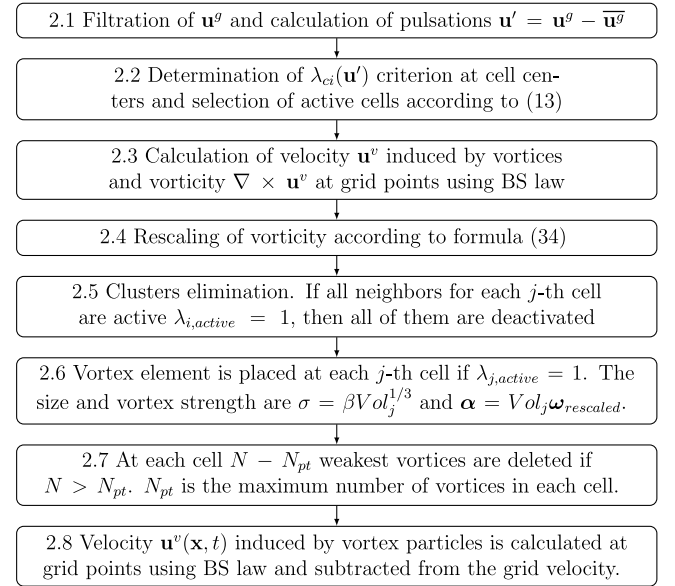


Fig. 6. Flowchart of the vortex particle identification procedure. BS stands for Bio-Savart..

- Redistribution.

The method was successfully tested in Fukuda and Kamemoto (2005) for the inclined collision of two vortex rings which is a popular test case for VPM. A disadvantage of this model is an avalanche-like increase of the vortex particles number due to the redistribution when the stretching occurs. The growth of vortex particles number should be counterbalanced by a special elimination procedure of Fukuda and Kamemoto (2005) based on a knowledge of a threshold for the dissipation rate which is difficult to set in a general flow case. To develop a robust code, to obtain a stable solution and to keep the particle number in a reasonable range we avoid the redistribution procedure in our computations. Thus, the finest vortices are removed. This reduces the range of scales that must be resolved in a numerical calculation. Such a reduction, as pointed by Chorin (1994), is a major goal of every turbulence model. From our side, we add that this is an immanent part of every turbulence model.

The particle displacement is calculated from the trajectory equation

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{u}_i^g + \mathbf{u}_i^v \quad (29)$$

The velocity is calculated from the Biot-Savart law taking into account adjacent vortex particles located only within one or two layers ($L = 2$) of neighboring cells. The simple Euler method is used for the integration of differential equations.

In the results presented below we use the vortex particles mollification of vorticity

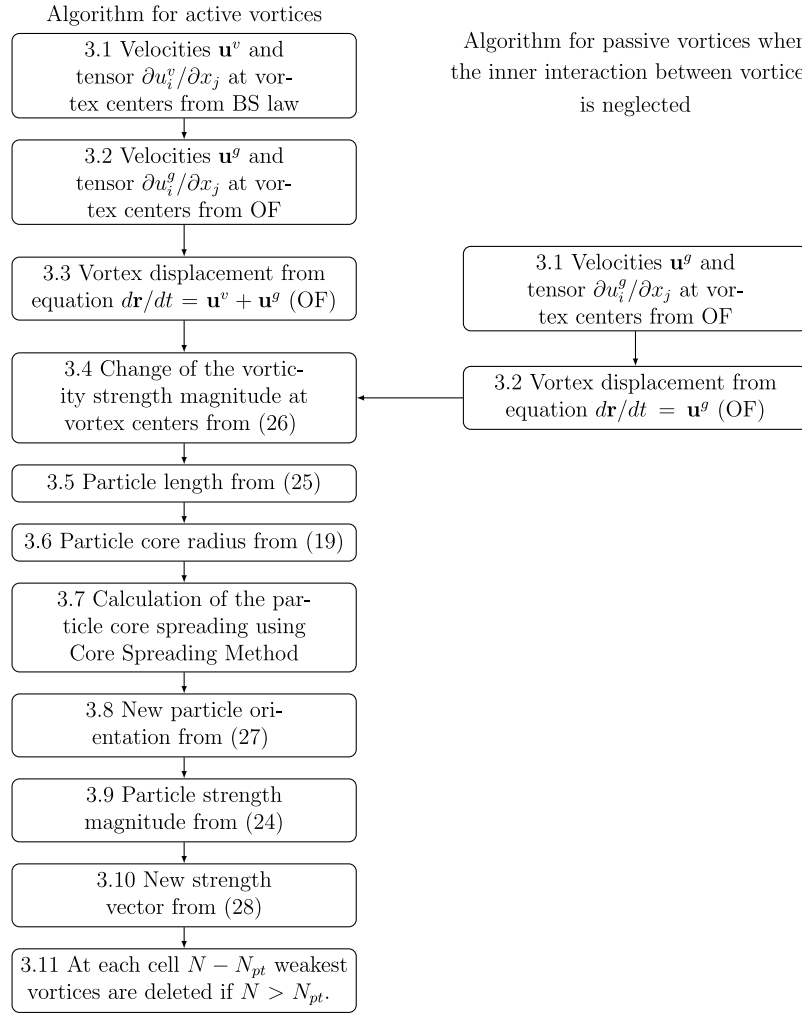


Fig. 7. Flowchart of the update procedure for the particle positions and strengths. OF stands for OpenFoam..

$$\omega = 3/(4\pi\sigma^3)\alpha e^{-\xi^3/\sigma^3} \quad (30)$$

proposed first by Moshier (1985). The velocity induced by such a particle and the strain rate tensor read

$$\mathbf{u}_p^v = \frac{1}{4\pi} \frac{\alpha \times \xi}{\xi^3} (1 - e^{-\xi^3/\sigma^3}) \quad (31)$$

$$\frac{\partial u_i^v}{\partial x_j} = \frac{3}{4\pi\sigma^3} \frac{\xi_j(\alpha \times \xi)}{\xi^5} \left(\xi^3 e^{-\xi^3/\sigma^3} + \sigma^3 (e^{-\xi^3/\sigma^3} - 1) \right) - \frac{1}{4\pi} \epsilon_{ijk} \alpha_k \frac{1 - e^{-\xi^3/\sigma^3}}{\xi^3} \quad (32)$$

The technique presented above is rather a framework which can be filled with any other more advanced procedures which can be implemented on arbitrary grids and efficiently parallelized.

3.4. Rescaling procedure

The computations showed that the velocity $\mathbf{u}^v$ is much less than $\mathbf{u}^v$ and not only because a part of vortex particles belonging to clusters were eliminated. Even if vortex particles would be generated at each cell the velocity calculated from the Bio-Savart formula $\mathbf{u}^v$ differs from $\mathbf{u}^v$. The reason is that the identity

$$\mathbf{u}^v(\mathbf{x}_j, \alpha) = \frac{1}{4\pi} \nabla \times \sum_{i=1}^M \frac{(\nabla \times \mathbf{u}^v(\mathbf{x}_i)) Vol_i}{|\mathbf{x}_i - \mathbf{x}_j|} = \frac{1}{4\pi} \nabla \times \sum_{i=1}^M \frac{\alpha_i}{|\mathbf{x}_i - \mathbf{x}_j|} \equiv \mathbf{u}^v(\mathbf{x}_j) \quad (33)$$

is satisfied in computations only in the limit $M \rightarrow \infty$. This means that

we have to use many vortex particles within each cell and sufficiently increase the computational costs. To avoid this and to get a proper level of $\mathbf{u}^v$, it is suggested to recalculate the strengths using the rescaling procedure

$$\alpha_i^{rescaled} = \frac{|\nabla \times \mathbf{u}^v(\mathbf{x}_i)|}{|\nabla \times \mathbf{u}^v(\mathbf{x}_i, \alpha)|} \alpha_i \quad (34)$$

3.5. Parallel implementation into openfoam

Implementation of the method into OpenFoam was sufficiently simplified by the availability of many routines developed in OpenFoam for particle laden flows. The class of particles was used for vortex particles. The routines for the interpolation of $\mathbf{u}^g$, $\partial u_i^g/\partial x_j$ and $\omega^g = \nabla \times \mathbf{u}^g$ from grid points to particles, particle tracking, filtering procedure to calculate $\overline{\mathbf{u}^v \times \omega^g}$, list of neighbors were available and effectively used to develop a fast working code. The flowchart of the whole algorithm is presented in Figs. 5, 6 and 7. The authors add the computation of the source term $\overline{\mathbf{u}^v \times \omega^g}$, the vortex particles identification routine and a routine for the solution of the vortex transport equation (7). The term $\overline{\mathbf{u}^v \times \omega^g}$ in (6) is treated in an explicit way. In the present code version most of routines were parallelized excepting computation of $\mathbf{u}^v$ and $\partial u_i^v/\partial x_j$ both at particle centers and grid points which need to be parallelized in the future.

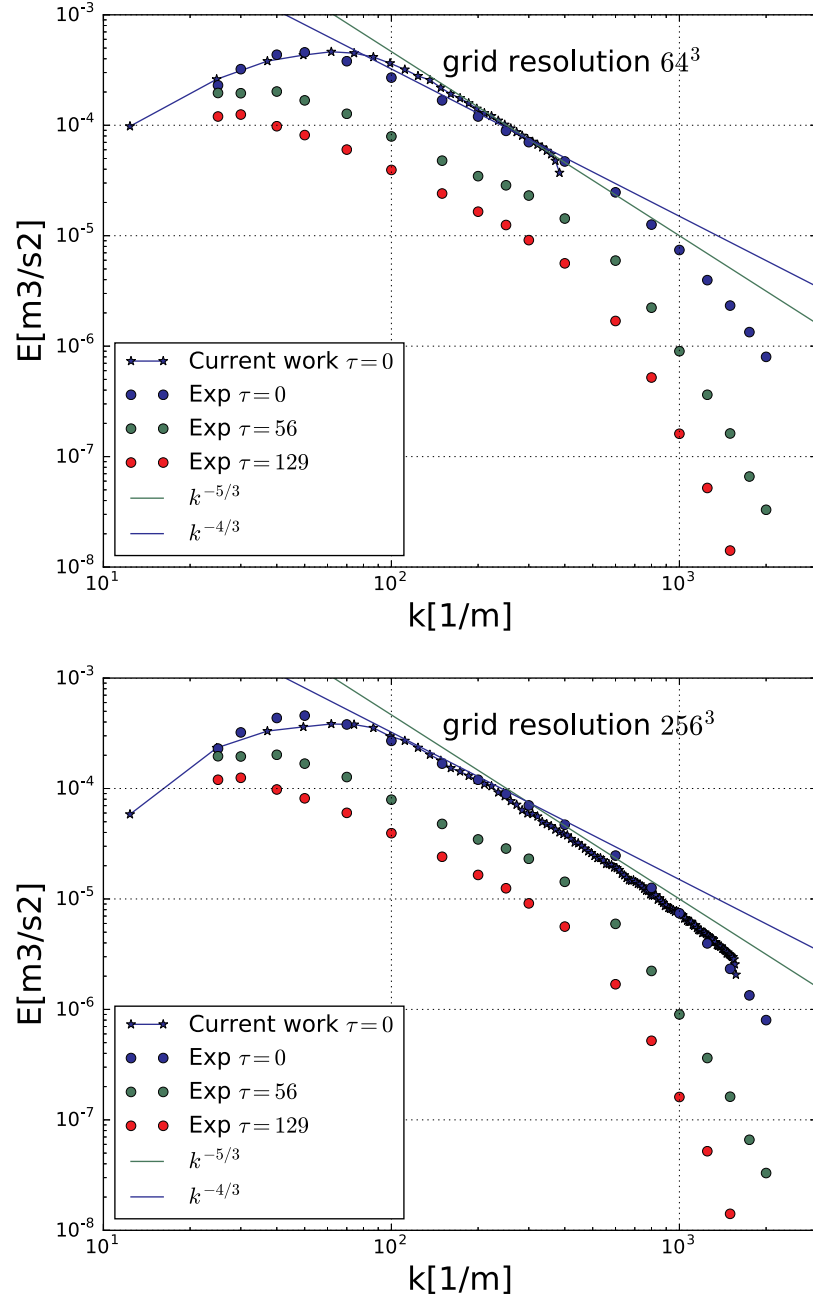


Fig. 8. Energy spectra of velocity field initialized by the inverse Fourier method.

3.6. Free parameters of the method

The method consists of a few free parameters which are:

- $\lambda_{ci, min}$ is a threshold to sort out weak fine scale vortices (see definition (13)),
- β is the overlapping ratio of vortex particles,
- γ is the parameter to map the large particles back to the grid. If a size of a vortex exceeds $\gamma\Delta$ it is mapped onto grid.
- N_{pt} is the permitted number of particles in cell.

The experience shows that λ_{ci} is high at the vortex and then abruptly decays. That is why the results are relatively insensitive to the choice of $\lambda_{ci, min}$ if this parameter is smaller than, say, ten percent of the maximum λ_{ci} determined in the whole flow. The overlapping parameter β is a typical parameter widely used in Vortex Particle Methods to attain a

proper accuracy and stability of computations. Unfortunately, it is difficult to find any unambiguous recommendation for its choice. In this paper we took $\beta = 2$ relying on the positive experience of Knio and Ghoniem (1990) and Feder et al. (2018). Parameter γ is taken four since all vortices with sizes larger than three cell sizes are not converted to the particles according to the algorithm described in Section 3.1. N_{pt} is the most critical free parameter which influence is proven in Section 4.6.

4. Results

The experimental data of Comte-Bellot and Corrsin (1971) are used as a well tried benchmark test for the decaying homogeneous isotropic turbulence which has been a popular test case in the evaluation of different turbulence models.

The numerical setup can be taken from numerous works. In this

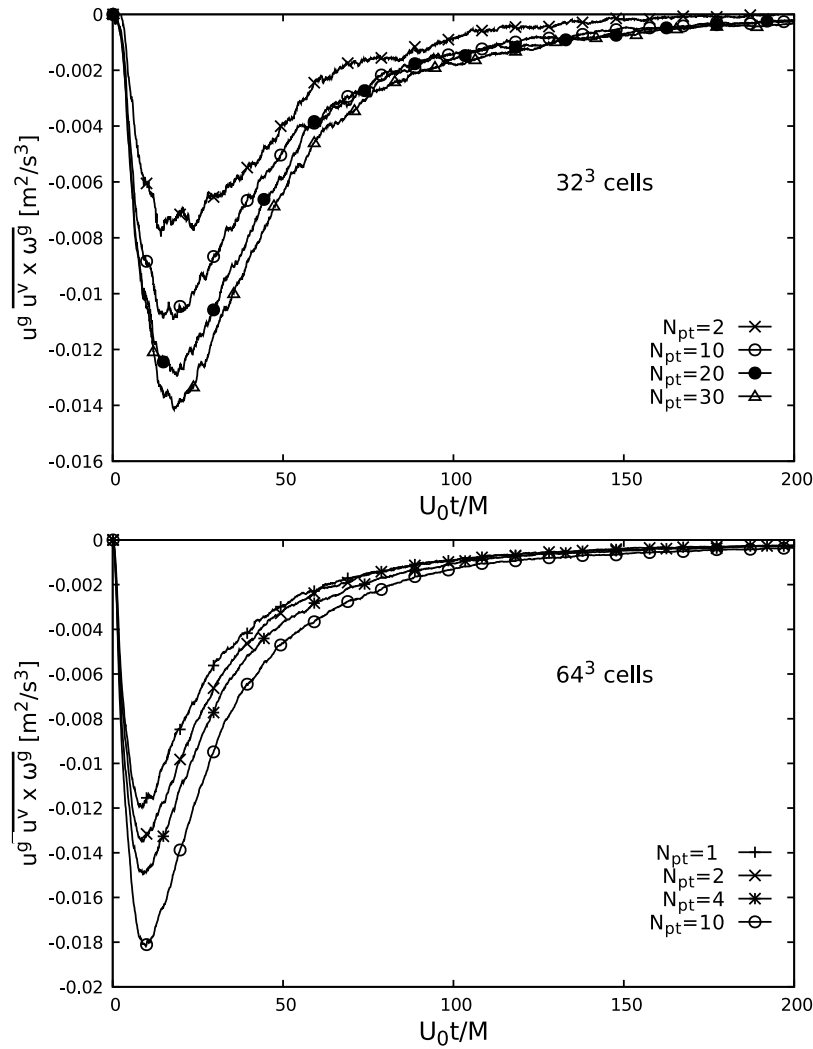


Fig. 9. Spatially averaged energy contribution of the term $\overline{\mathbf{u}^v \times \omega^g}$.

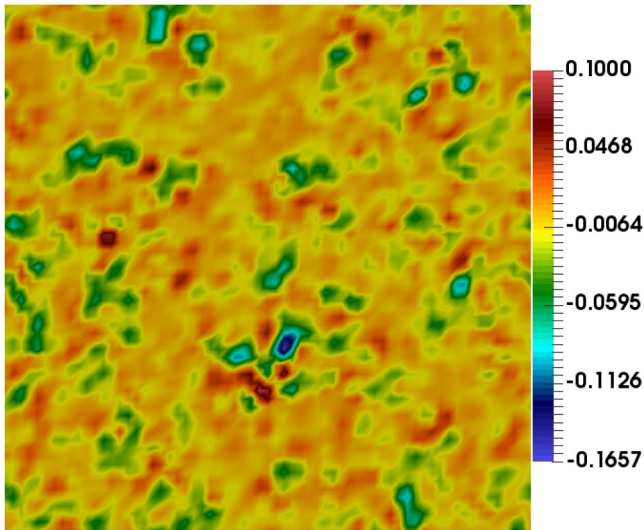


Fig. 10. Instantaneous distribution of the energy production term $\mathbf{u}^g \mathbf{u}^v \times \omega^g$ in the middle plane of the computational box at $\tau = 56$. Positive values correspond to the energy increase. 64^3 grid.

paper we used the setup proposed in Dietzel et al. (2014) and Sohm (2007). The computational domain in form of a cube with the size $L = 0.508\text{m}$ was meshed with uniform Cartesian grids with 32^3 , 64^3 , 128^3 and 256^3 cells. The periodic boundary condition was enforced on each face of the cubical box. The Reynolds number based on the Taylor micro scale, air kinematic viscosity $\nu = 1.5 \cdot 10^{-5} \text{m}^2/\text{s}$ and the root mean square of the longitudinal velocity $u_{rms} = 0.222 \text{m/s}$ is $Re_\lambda = u_{rms} \lambda / \nu = 71.6$. The results are evaluated at three dimensionless times of $\tau = 0$, $\tau = 56$ and $\tau = 129$ which, with employment of the Taylor hypothesis, correspond to three different locations downstream of the grid at $\frac{U_0}{M} = 42$, 98 and 171. Here $M = 0.0508 \text{m}$ is the size of the grid, which was used for generation of turbulence in the experiments, U_0 is the air stream speed and t is elapsed time in travelling at the mean flow velocity from the grid. Following to Dietzel et al. (2014), the initial velocity field was generated artificially using the Inverse-Fourier approach based on a very popular technique proposed by Kraichnan (1970). In this work, further developments taken from Billson and Davidson (2003) and Saad et al. (2016) were considered for implementation in OpenFOAM. The synthesis of the turbulence was performed for the Karman-Pao spectrum using 5000 modes (Saad et al. (2016)). The integral length $L = 0.024 \text{m}$ is equal to that measured by Comte-Bellot and Corrsin (1971). Quality of the initial turbulence field is illustrated in Fig. 8 showing the convergence of the spectrum to the target one when the resolution increases from 64^3 to 256^3 cells.

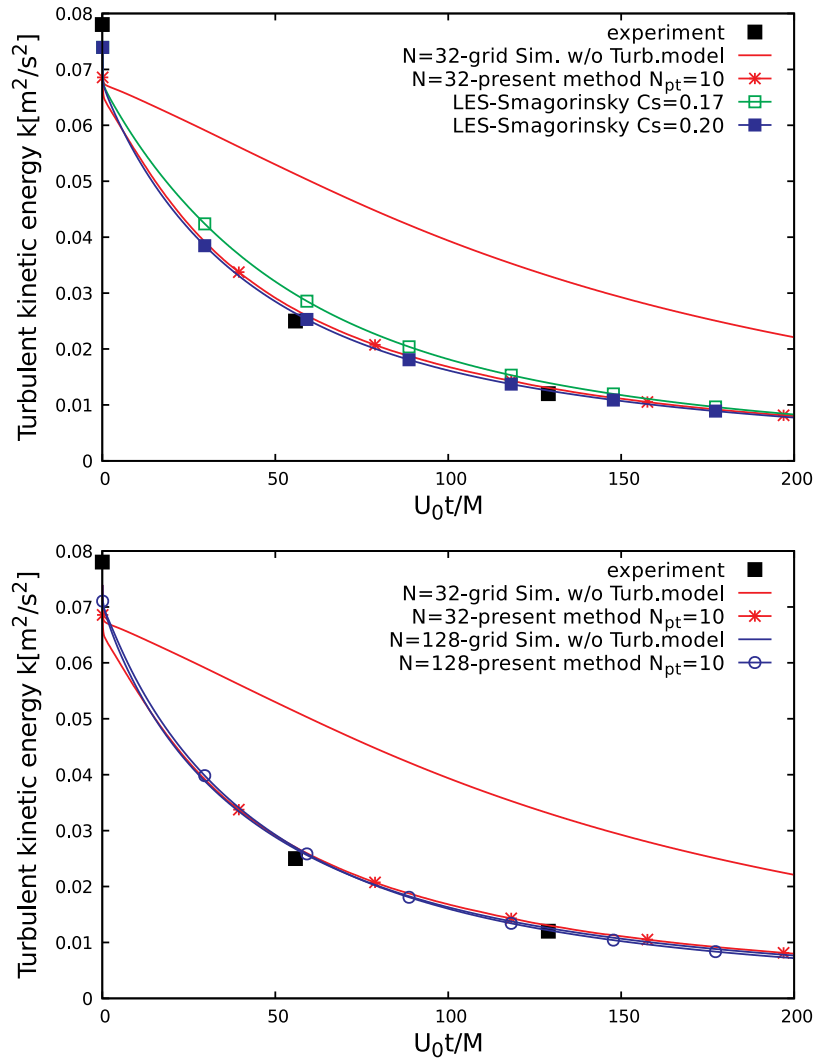


Fig. 11. Change of the kinetic energy $k = (u_i^g + u_i^v)^2/2$ averaged on the box versus time.

For time discretization a second order Crank-Nicolson scheme was used. The spatial discretization of the convective term was performed using the second order central difference scheme. The CFL criterion was kept less than 0.1 according to recommendations of [Sohm \(2007\)](#). Simulations were carried out using the OpenFoam code ([Weller et al. \(1998\)](#)). The simulation of the Eqs. (6) and (7) is performed with the same time step although the large and fine scales should have different time scales. As shown below in [Section 4.7](#), the interaction between vortex particles, which is expected to be fast, plays a negligible role at least for the test case under consideration. Therefore, the assumption of the single time step can be considered as acceptable one.

4.1. Contribution of the term $\overline{\mathbf{u}^v \times \omega^g}$ to the energy budget

Multiplying the equation (6) with u_i^g we obtain the equation for the kinetic energy of grid based velocity field $k^g = \frac{1}{2} u_j^g u_j^g$:

$$\frac{\partial k^g}{\partial t} + u_j^g \frac{\partial k^g}{\partial x_j} = \frac{\partial}{\partial x_j} \left(P^g u_j^g + \nu \frac{\partial k^g}{\partial x_j} \right) + \mathbf{u}^g \overline{\mathbf{u}^v \times \omega^g} \quad (35)$$

The last term in (35) describes the contribution of the fine scales to the energy of the grid based motion. [Fig. 9](#) illustrates the temporal change of this term averaged on the whole computational box. A very important conclusion is that the term $\overline{\mathbf{u}^v \times \omega^g}$ behaves as an energy drain transferring the energy of the grid based motion into fine scale energy.

With the other words, it acts as a diffusive Large Eddy Simulation subgrid model. This is true only in the averaged sense because in contrast to pure diffusive models like the Smagorinsky one, the present term allows the energy back scattering, i.e. flux of the energy from fine scales to the large ones. Indeed, as mentioned above, this term reproduces amplification of the grid based vorticity in field of $\mathbf{u}^v$ and can lead to a local increase of the kinetic energy. This is illustrated in [Fig. 10](#). [Fig. 9](#) shows that the energy drain gets larger when the number of particle per cell N_{pt} (see definition in [Section 3.1](#)) increases. This is a quite logic result, because the energy of fine scale motion grows when N_{pt} increases. The flux of energy, discussed above, is only a part of the total flux. Another part, which is definitely negative, is due to conversion of grid based vorticity into vortex elements.

4.2. LES Like behavior and consistency of the model

As mentioned above, the coupling term has the same property as the LES subgrid stress with respect to the energy budget. [Fig. 11](#) is an additional confirmation of this thesis. For coarse resolution with 32^3 cells, the kinetic energy $k = (u_i^g + u_i^v)(u_i^g + u_i^v)/2$ averaged on the box decays like in LES simulations performed with a simple Smagorinsky model with $C_s = const$ and with the same resolution as in hybrid embedded simulations. The pure grid simulations performed with OpenFoam without any turbulence model indicate a delayed underestimated energy decay (red line). The lower picture illustrates a remarkable fact.

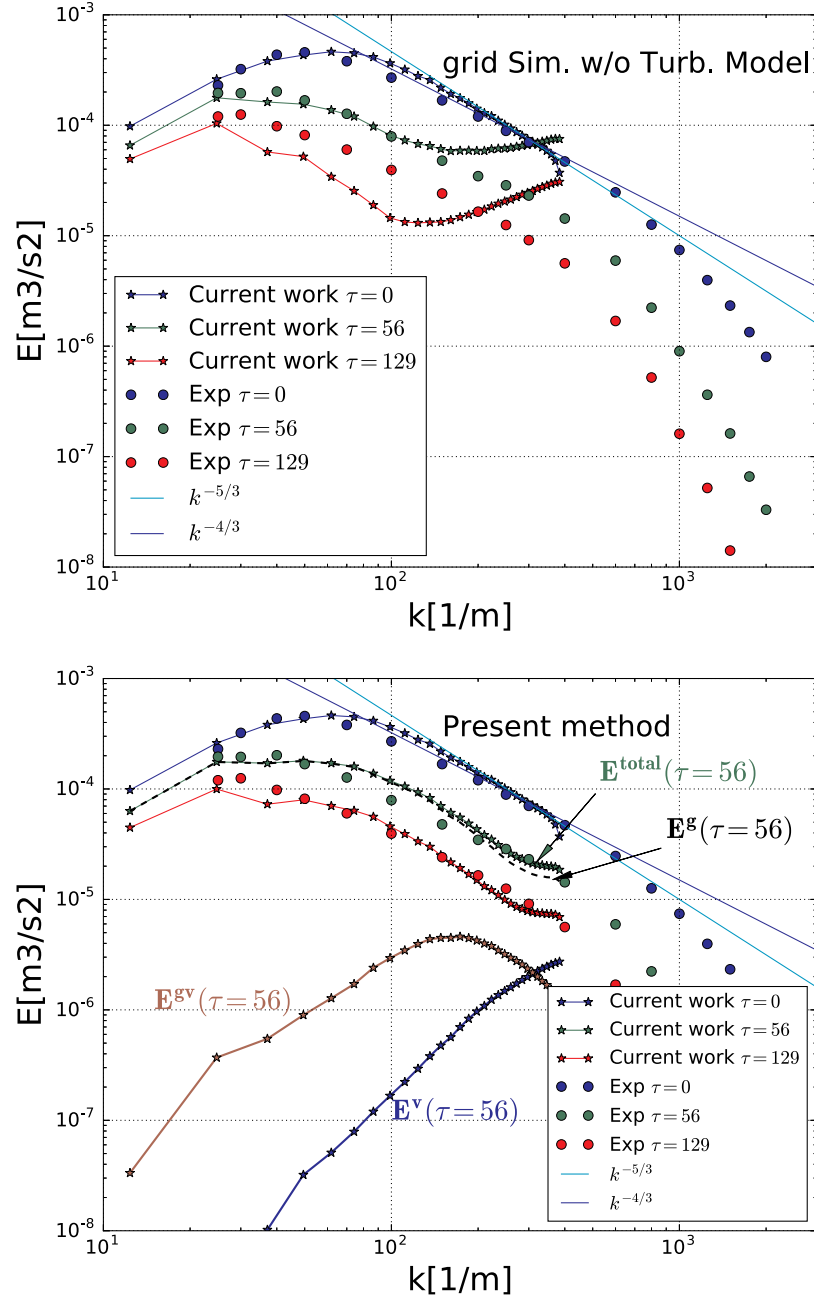


Fig. 12. Spectrum E obtained using three different methods with 64^3 grid. $E^g = (u^g)^2/2$, $E^v = (u^v)^2/2$ and $E^{gv} = E - E^g - E^v$ are different contributions to the spectrum (middle picture).

When the resolution grows, the effect of the term $\overline{\mathbf{u}^v \times \omega^g}$ reduces, the present method converges to the purely grid simulation (compare results for 32^3 and 128^3 cells grids). With the other words, the present method is consistent and converges to the Direct Numerical Simulation when the resolution increases. In all simulations there is a clear drop of the kinetic energy at the first time step reported also by Dietzel et al. (2014). When the resolution grows the drop slightly decreases but not sufficiently. Our study showed that this is an artefact which is specific for the method of artificial generation of initial turbulence used in this paper.

4.3. Influence on the spectrum and autocorrelation function.

The energy at high wave numbers is overpredicted in purely grid simulations without turbulence model (see Fig. 12). Since there is no drain of energy it is accumulated at small structures. This can also be

considered as a sign of numerical instability. Strong drop of the autocorrelation function at $r \sim 0$ (see red curve in Fig. 13, top) points out on the generation of small structures with scales proportional to one cell size. The vorticity pattern becomes very spotty. Due to the drain of energy by the term $\overline{\mathbf{u}^v \times \omega^g}$ the grid solution becomes more stable and energy content at high wave numbers is sufficiently reduced. The form and the temporal change of the autocorrelation function becomes more relevant which widening indicates the increase of the integral length. However, a certain pile-up of E remains at large wave numbers indicating a sort of the numerical instability. Fig. 12, middle explains this effect. The energy of small scales vortex particles E^v grows with k when their size decreases due to the stretching. The problems with spectrum at high wave numbers completely disappears at high resolution grid with 256^3 cells. When the resolution grows, spectra obtained using the present method converge to those of the purely grid simulations demonstrating the consistency of the method (not shown in Fig.).

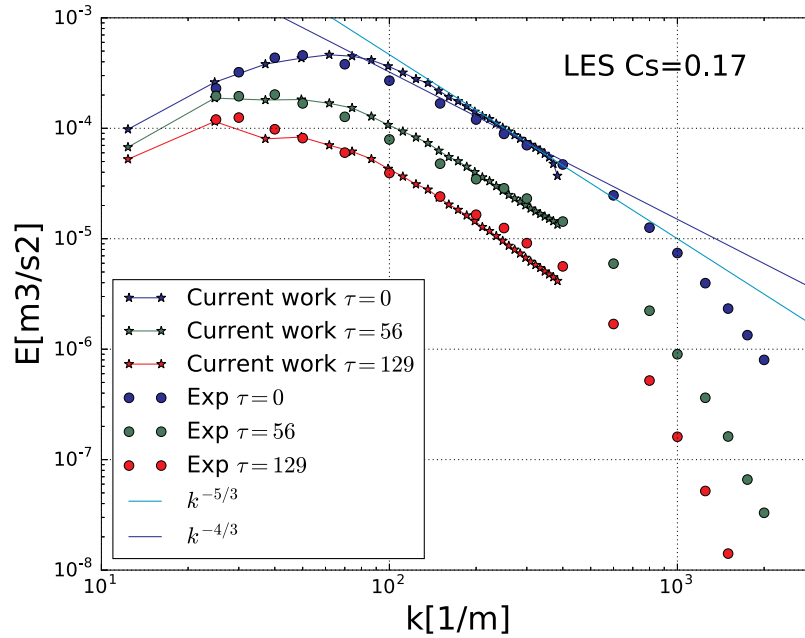


Fig. 12. (continued)

4.4. Capturing fine scales

Probability density function of the vortex sizes σ is presented in Fig. 14 at different time steps. As seen, there is a big variety of fine vortices with sizes ranging from 0.1Δ to Δ . This is an illustration of the fact that the present method is capable of resolving scales an order less than the grid cell size even at small $N_{pt} = 4$. The vortices smaller than 0.1Δ are excluded by selection of strongest vortices in cells and restriction of N_{pt} . In principle, they could be much smaller. The only restriction is the computational time and memory space. Resolution of the vortices with $\sigma \sim 0.1\Delta$ in pure grid methods would require grids with number of cells at least 10^4 larger than in the embedded simulations provided each vortex would be represented only by three cells. The largest vortex size of 4Δ is a specific feature of the present algorithm, because if $\sigma > 4\Delta$ the vortex particle is mapped back to the grid. The p.d.f. distribution is shifted towards larger σ in time. This tendency is due to temporal growth of the structures in the grid based flow which is in accordance with widening of the autocorrelation function (Fig. 13) and increase of the integral length.

4.5. Simulation of mixing

Previous results showed the consistency, LES-like behavior of the method and reliability of results for statistical parameters of velocity fluctuations field. A potential of the method and its advantage over common LES methods and purely grid simulations is demonstrated in this subsection for the scalar field dynamics which is modeled by the scalar transport equation

$$\frac{\partial \phi}{\partial t} + (\mathbf{u}^v + \mathbf{u}^g)_j \frac{\partial \phi}{\partial x_j} = D \frac{\partial^2 \phi}{\partial x_j \partial x_j}, \quad (36)$$

where ϕ is the total passive scalar consisting both large and small scales. Since (36) doesn't contain unclosed terms the concept of the turbulent Schmidt number is not necessary. In other words, this simulation is a kind of DNS for scalar.

The initial scalar field is configured in a way that half of the computational domain has the mixture fraction $\phi = 1$ and the rest is $\phi = 0$. The rate of the scalar mixing is described by the scalar dissipation rate

$$\chi = 2D(\nabla \phi)^2$$

where D is the diffusion coefficient taken from the condition that the Schmidt number is 1.0.

Fig. 15 shows the evolution of spatially averaged scalar dissipation rate for the grid resolution 64^3 . As can be seen, compared with the case of pure grid simulations without any turbulence model, the present method remarkably increases the scalar dissipation rate. Obviously, the small-scale vortices which are retrieved and preserved in embedded simulation approach intensify the scalar gradient. Additional perturbations of the velocity field by fine vortices, detected only at grid points, are proved to be sufficient to substantially increase the scalar dissipation rate χ . The LES results were obtained using the gradient-diffusion hypothesis with the turbulent Schmidt number of $Sc_t = 0.7$. The resolved part of $2D(\nabla \phi)^2$ is relatively low for the coarse resolution 64^3 . To be compared properly the subgrid contributions $2D(2\partial\tilde{\phi}/\partial x_i \partial\phi'/\partial x_i + \partial\phi'/\partial x_i \partial\phi'/\partial x_i)$ should be extra added and numerical results are to be compared in a filtered form.

Strictly speaking, for the consistent analysis the scalar dynamics should be modeled at $Sc \sim 1$ with the same resolution as the velocity field. In principle, it can be done within the formalism of the particle based techniques. For details the reader is referred to the recent papers of Lagaert et al. (2012) and Matsunaga et al. (2015). Since it requires additional serious development and could be a subject for further study, in this paper the scalar dynamics is resolved on the grid to demonstrate that χ increases when the fine scales of the velocity field are resolved. With other words, the aim of this subsection was to show the potential of the method to solve the problem of underestimation of the mixing in LES simulations.

4.6. Influence of N_{pt}

One of the weaknesses of particle methods is the presence of free parameters which are often tuned to match desirable results. Strictly speaking, the number of particle per cell N_{pt} is not a freely selectable parameter within the present method. It should be as large as possible. Fig. 16 shows the convergence of numerical results for k and χ when N_{pt} increases. The spatially averaged kinetic energy is non sensible to N_{pt} . For χ the value and place of the peak is independent of the number of particles and a slightly higher mixing rate is achieved for a longer time of simulation for higher number of particles. Such a saturation indicates that the numerical results become independent of N_{pt} after some relatively small threshold value $N_{pt} \sim 4$.

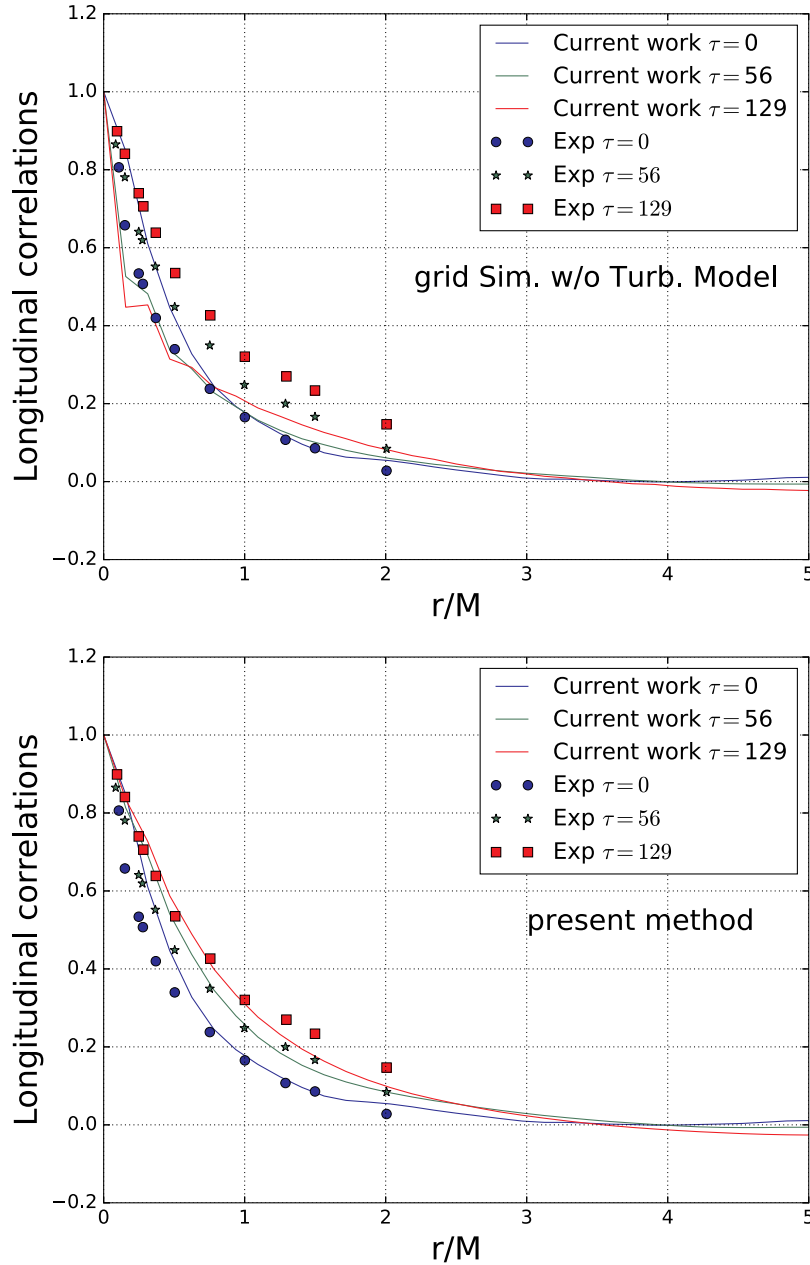


Fig. 13. Longitudinal autocorrelation coefficient obtained using three different methods with 64^3 grid.

4.7. Influence of the inner interaction between vortex particles.

As mentioned above, laboriousness of calculation of the velocity induced by vortex particles is a well known limitation of the computational vortex particle method because the number of required operations is proportional to $O(N^2)$. Within the present approach the calculation of $\partial u_i^v / \partial x_j$ and the velocity induced by particle on each other $\mathbf{u}_p^v$ (see expressions (31) and (32)) is the highest time consuming part of calculations with $O(L^2 N_{pl}^2)$ operations, where L is the number of adjacent neighboring cells. Thus, it is necessary to estimate the importance of these calculations on the spatially averaged results. Fig. 17 shows results obtained when the inner interaction between particles is neglected and they are driven only by the grid based flow. With this assumption the equation (7) takes the form

$$\frac{d\omega^v}{dt} = (\omega^v \cdot \nabla) \mathbf{u}^g + \nu \Delta \omega^v, \quad (37)$$

and the r.h.s of the trajectory equation (38) contains only the grid based

velocity

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{u}_i^g \quad (38)$$

As seen, the influence of inner interactions on spatially averaged kinetic energy and scalar dissipation rate is relatively weak and can be neglected in the calculation. Thereby the computations can be done sufficiently faster. It seems to be that the vortex particles serve just as triggers of turbulence and their inner interaction doesn't contribute sufficiently to the flow evolution. This issue should thoroughly be studied in future works for more complex flows. The insignificance of the interaction reduces the importance of the increase of the scheme accuracy for the integration of the particle trajectory and particle strength equations. The simple Euler method can be used as the most simple and fast procedure.

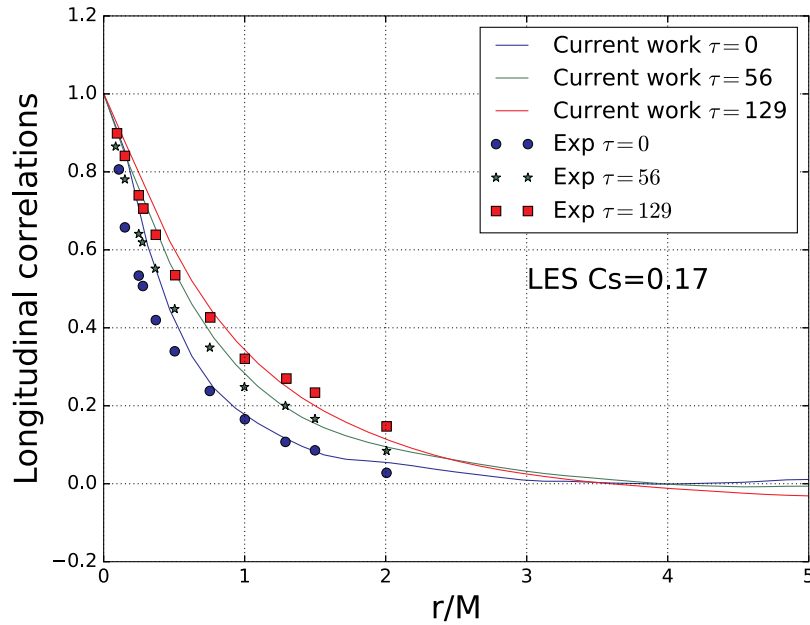
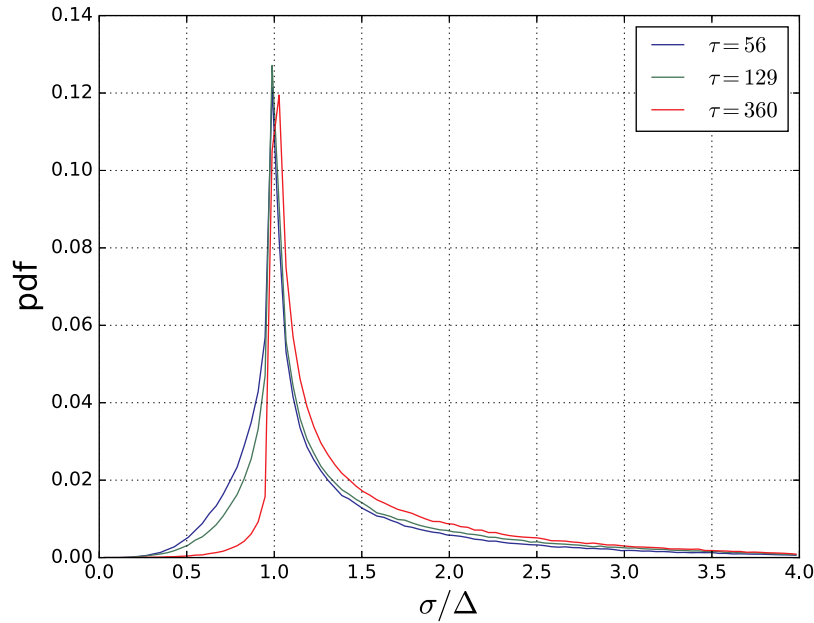


Fig. 13. (continued)

Fig. 14. P.d.f of the vortex size σ for the grid 64^3 and $N_{pt} = 4$.

4.8. Estimation of the computational time.

Since a part of our code implemented into OpenFoam has still not been parallelized, the computational time study was performed for the one-processor computations. For the same grid, the resolution of the subgrid motion proposed in the present method increases the computational time in comparison with LES sufficiently. For instance, the computational time of one time step on the grid with 32^3 points is

- 0.123 s for the simulation without turbulence model,
- 0.14 s for the LES simulation with Smagorinsky model,
- 0.563 s for the hybrid simulation with passive vortons (without inner interaction) and two particles per cell $N_{pt} = 2$.

The most time consuming operations are those requiring application of the Bio-Savart summation and data interpolation from the grid to

particles: steps 2.3, 2.4, 2.8, 3.2, 3.3 and 3.4 in flowchart Fig. 6 and 7. They require totally 0.39 s or about 89% of the additional computational time. These results look at first glance, perhaps, pessimistic. However, one has to bear in mind the following considerations. First, resolution of additional scales requires additional computational costs. As shown in Fig. 14 for the grid 64^3 we resolved structures with sizes ten times less than the cell size, i.e. 0.1Δ . Resolution of these structures using a pure grid method using only four cells per structure would require $64^3 \cdot ((\Delta/(0.1\Delta))^3 \cdot 4^3) = 1.7 \cdot 10^{10}$ cells.

Second, and this is a very important note, one has to compare the computational costs of two simulations with the same level of accuracy. For instance, in our recent jet simulations (Samarbakshsh and Kornev (2018)) the accuracy of our approach with $1.5 \cdot 10^5$ cells is the same as that of LES with 6Mio cells, whereas the LES with the same resolution is fully improper. Also the LES computations with 3Mio and 5Mio grids shows unsatisfactory comparison with measurements. The

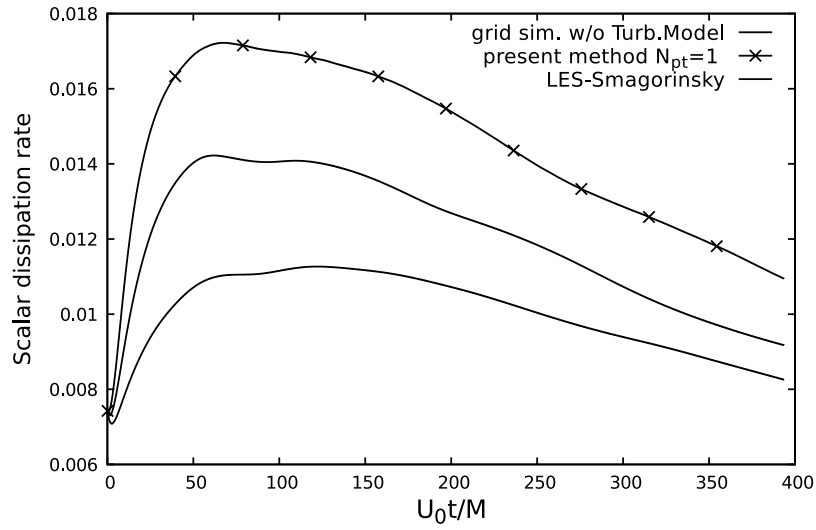


Fig. 15. Evolution of the spatially averaged scalar dissipation rate χ for the grid 64^3 . LES (Smagorinsky SGS model with the constant $C_s = 0.17$) results were obtained using only the resolved gradients.

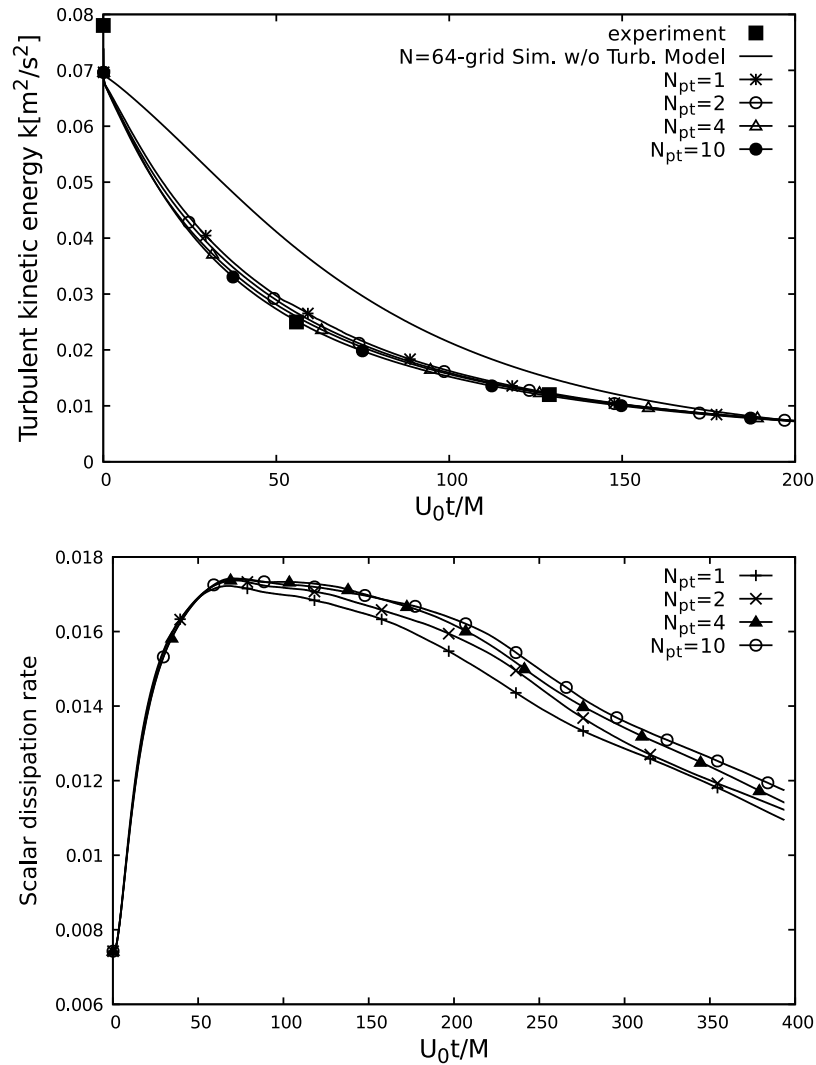


Fig. 16. Influence of the parameter N_{pt} on the spatially averaged kinetic energy (top) and scalar dissipation rate (lower) for the grid 64^3 .

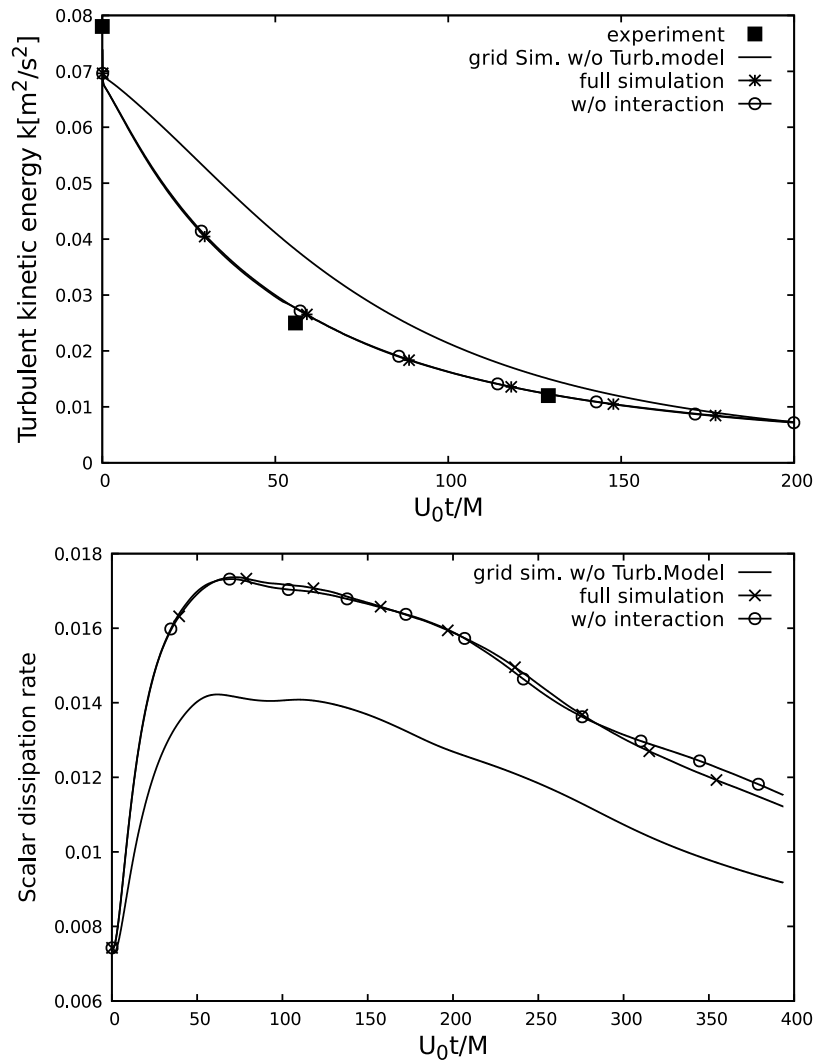


Fig. 17. Influence of interaction between vortex particles on the spatially averaged kinetic energy and scalar dissipation rate for the grid 64^3 .

computational time for one time step using LES with 6 Mio is 65 s whereas the present method requires only 3.2 s, i.e. the speed up factor is around 20. Certainly, the parallelization will change this ratio because the finite volume method and VPM have different parallelization efficiency.

5. Conclusions

A new method for simulation of multiscale turbulent flows, recently proposed and tested for two dimensional flows in Kornev (2018), is applied in this work for a three dimensional case. The method is based on the decomposition of the flow structures in large scale ones, resolved on the grid, and small scale ones, represented by vortex particles. The evolution of both scales is described by a set of two transport equations which contain the terms describing the coupling between the scales. The method is simple, straightforward and utilizes advantages of grid based and grid free simulations. Since the vortex particles are not influenced by artificial viscosity, typical for grid methods, they can survive in the solution for relatively coarse grids. Thus, a relatively high resolution of the flow can be attained at coarse grids. Although the theoretical basis of the method has been described in Kornev (2018), its application to three dimensional flows required recapitulation of the algorithm with respect to the vortex particle generation procedure, computation of the particle induced velocities and numerical solution of the vorticity transport equation. The modifications, presented in this

paper, were also sufficiently dictated by peculiarities of the incorporation of the final algorithm into the framework of the OpenFoam code for big unstructured grids of arbitrary complexity.

This paper presents the results of validation of the method for a well tried benchmark test of the decaying homogeneous isotropic turbulence using the experimental data of Comte-Bellot and Corrsin (1971). A very important conclusion of validation is that the spatially averaged term $\overline{\mathbf{u}^v \times \omega^g}$, which describes the influence of fine scales on large ones, behaves as an energy drain transferring the energy of the grid based motion into the fine scales energy. At coarse resolutions, it acts as a diffusive Large Eddy Simulation subgrid model resulting in a LES-like behavior of the whole method. When the resolution increases, the present method is consistent and converges to the Direct Numerical Simulation. It was shown that the present method is capable of resolving scales much less than the grid cell size. In such a way, very fine scales can be resolved on coarse grids. The plausibility of simulations were successfully checked for the second order statistics, i.e. for the spectral density E and the longitudinal autocorrelation coefficient. An important advantage of the method over common LES methods and purely grid simulations is demonstrated for the scalar field dynamics at the Schmidt number of 1.0. Compared with the case of purely grid simulations without any turbulence model, the present method remarkably increases the scalar dissipation rate, because the small-scale vortices which are retrieved and preserved in embedded simulation approach intensify the scalar gradient. The methodical study

demonstrated that the inner interaction between particles can be neglected at least for the test case considered in this paper. This results in a very efficient and fast computational procedure.

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